`ProxyFinder` : Proxy Selection in Divergent Data Sets by Methodological Prediction and Correlation Reduction

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ABSTRACT

Although the problem of missing data has been studied extensively within the imputation literature, a more vexing problem has long plagued social science research: finding a suitable proxy for an entirely unobserved variable. Indeed, contemporary proxy selection methods include the manual analysis of (often hundreds of) survey items, variable codings, and variable distributions to identify a variable that exhibits face-validity. Our novel algorithm, ProxyFinder, aims to automatically identify a proxy variable in one (testing) dataset by analyzing in a separate (training) dataset the variable that we would like to approximate. ProxyFinder works in three stages: prediction, correlation reduction, and proxy selection. The first stage relies on existing imputation methods to generate predictions for the missing variable in the test dataset. The second stage calculates a score for each potential proxy variable that balances the desiderata of convergent and discriminant validity. In the final stage, ProxyFinder selects the candidate variables with the highest proxy scores.

Keywords: Missing Data, Imputation, Proxy Variable

Code available at `https://github.com/Kirin2D2/ProxyFinder`

1 INTRODUCTION

It is common for large datasets in the social sciences to omit a variable of analytical interest to researchers. Since it is often infeasible and expensive to re-create or augment datasets—imagine tracking down 10,000 respondents in an opt-in survey to ask an additional question—researchers often focus on identifying a stand-in proxy variable. These proxy variables are measured variables that aim to accurately explain the variance of the unobserved target for use in statistical analysis.

Conventional solutions to the problem of missing targets generally involve either choosing a single variable that is theoretically face-valid for the target variable or using methods such as factor analysis or regression to create a synthetic proxy from multiple observed variables. Unfortunately, datasets commonly contain hundreds of variables, and it can be time-consuming for experts to manually analyze variables to decide which ones (if any) capture the underlying concept. A more pressing issue arises when there is no clear face-valid proxy available. In this case it is difficult to ensure that a variable captures only (or even primarily) the construct variance, and not the variance of a confounder. Synthetic proxies, on the other hand, can be useful, but only when a large portion of the target variable's variance is explained by the measured variables. Of course factor analysis and regression cannot be used if the dataset is missing the target (unless it is run on a different, complete dataset with candidate proxies that are in both datasets, and assumed to generalize) (Franklin, 1989). When a missing target variable does not have a well-established proxy, experts spend tedious hours searching for face-valid proxies by analyzing distributions and relations to other variables (Hobbs and Vignoles, 2010; Rapeli, 2022).

We propose an automated proxy selection algorithm called "ProxyFinder" to more effectively identify a suitable proxy for a given target variable. We introduce a 'penalization weight' to balance representation of the target construct with non-representation of confounding predictors. Initial testing has shown that ProxyFinder outperforms synthetic proxy generation via two standard neural network-based imputation methods in selecting a measured variable that maintains both convergent and discriminant validity. It is also able to identify reasonable proxies when no clear face-valid options are available.

We suspect that the optimal correlation penalty (described below) and the quality of the proxy selection process depend on the accuracy of the model's predictions. Employing other imputation methods (e.g., classification models) could improve performance, especially for categorical and discrete variables which dominate social science datasets. Regardless, our testing has shown that ProxyFinder selects strong proxies for specific targets such as the favorability of Donald Trump and Joe Biden, both within single and divergent datasets.

1.1 Background

Social scientists have long used substitute variables to analyze underlying concepts, but formal discussion of proxying latent variables emerged in the last 30 years (Kmenta, 1991). According to Graham Upton's *A Dictionary of Statistics*, a proxy variable is a "measurable variable that is used in place of a variable that cannot be measured" (Zahur, 2009; Upton and Cook, 2008). One or more proxy variables are commonly used by social scientists to make inferences about variable(s) of interest or to conduct large-scale data analysis (Zahur, 2009). A good proxy should demonstrate face validity, construct validity, and discriminant validity (i.e. capture as much construct variance while minimizing variance from confounding variables) (Rapeli, 2022). A proxy is face-valid if it is theoretically judged by the researcher (a content expert) to represent the same underlying construct as the target (Rapeli, 2022). Construct validity can be divided into three subtypes: convergent, criterion, and predictive validity (Henseler, Ringle and Sarstedt, 2015; Rapeli, 2022). A proxy exhibits convergent validity if the proxy and target variables represent the same underlying information (latent construct). A proxy exhibits criterion validity if it is associated with other variables that we would expect it to be associated with given knowledge of the target. A proxy exhibits predictive validity if it can be used to accurately predict the target. Finally, discriminant validity can be understood as the extent to which a proxy is empirically unique; i.e., that it does not represent the same underlying information as other (unrelated) variables, and hence is not a proxy for some other concept(s) (Henseler, Ringle and Sarstedt, 2015).

In social science research, it is common that a researcher would like to analyze a concept that is not directly measured in a dataset. Measuring an excluded concept can be achieved in the following ways: First, if extant research can be used to assert the validity of a proxy for the missing concept and the proxy exists in the dataset of interest, then the researcher can simply use the known proxy. If, however, no prior proxy-related research has been conducted, the researcher is faced with a situation that is complex and tedious at best, and project-ending at worst. Although computational methods exist to help researchers find a proxy when the target variable is directly measured, to the best of our knowledge there is no accepted algorithmic method to select a proxy for a target variable that is not measured in a dataset (Zahur, 2009). Indeed, the likely strategy would be to manually check each available variable for face and construct validity until one meets the desired criteria.

1.2 An Automated Approach

Our initial thought was to expedite the proxy selection process by building upon research conducted by Franklin (1989). The method Two-Stage Auxiliary Instrumental Variables (2SAIV) infers a relationship between two variables that are measured in separate (i.e., no respondent overlap) datasets by extracting information from a set of variables that are observed in both datasets. Let x_d denote explanatory terms, y_d

denote outcomes, $d \in \{1,2\}$ denote datasets, and z denote predictor covariates. Consider a case in which we'd like to compare the variable x_1 in the first dataset with the variable y_2 in the second dataset (and neither variable is observed in both datasets). The method works via two OLS regressions. The first learns the equation $x_1 = \mathbf{z_1}\gamma + \varepsilon_1$, and then uses this to create $\hat{x_2} = \mathbf{z_2}\hat{\gamma}$ in the second dataset. The second simply estimates $y_2 = \hat{x_2}\beta + u$.

We attempted to use this method to generate a synthetic proxy in the second dataset, to approximate a target variable x that is observed in the first dataset and unobserved in the second. To test this method, we generated a synthetic proxy in one dataset where x was observed. Specifically, we split the dataset with an 80-20 train-test split, and treated the training set as a first dataset and the testing set as a second. We then used the first OLS regression from 2SAIV (i.e., $\hat{x_2} = \mathbf{z_2}\hat{\gamma}$) to generate synthetic values for x_2. We compared these synthetic values $\hat{x_2}$ with the observed values for x_2. Unfortunately, synthetic proxies tended to have higher correlations with the variables used in proxy generation than they did with the observed target, hence violating discriminant validity.

We next attempted to improve upon this technique by treating it more broadly as a univariate missing data problem. This approach used state-of-the-art imputation methods to generate synthetic variables in the dataset of interest to be used as proxies by training the imputation model on a separate dataset that contained both the target variable and a set of predictor variables present in both datasets. However, we quickly ran into the same problem of the synthetic proxies having stronger correlations with the predictor variables rather than the target variable which we sought to impute. While this approach exhibited reasonable convergent validity, it violated both face validity and discriminant validity, even with the addition of regularization, simulated random sampling, and the training of a discriminator.

We theorize that our initial attempts failed because of two mutually exclusive problems. In some cases, the synthetic proxy was not able to sufficiently capture the variance of the target, owing to high levels of error from the model's predictions. In other cases (or for other targets and/or predictors), relations between the predictor variables were too strong to differentiate the synthetic proxy from the predictor variables. We focus on the latter problem– known within the field of Machine Learning as "Shortcut Learning," i.e., the model learning a spurious "shortcut" between one or more predictor covariates and the target variable. To address this issue, we introduce a "correlation penalty" (which we elaborate on below). The corresponding algorithm (i.e., ProxyFinder) utilizes a new objective in which we reward a candidate proxy's ability to explain the variance of the target (i.e., convergent validity) and penalize its ability to explain the variance from the confounding variables (i.e., discriminant validity). The basic algorithm has the user input a training and testing dataset, the target variable, and a set of variables that can be used to predict the target in both datasets. It outputs a list of top-recommended proxies. ProxyFinder cannot

infer face validity; nor can it find a reasonable proxy if none exists in the testing dataset. Therefore, it's crucial that content experts independently evaluate the validity of the recommended proxies. The benefit of ProxyFinder is to identify the best candidate proxies when perfect ones may not be available, and to formalize the selection process in a way that considers both convergent and discriminant validity.

2 METHODS

2.1 Definitions

DS1: Dataset 1 contains the target variable and the predictor variables we use to predict the target. This dataset is used for training an imputation model for the target variable.

DS2: Dataset 2 contains the variables that we would like to consider as candidate proxies for the target and the predictor variables we will use to impute the target.

t: The target variable is the column in DS1 that we would like to find a proxy for in DS2.

v: A variable.

$\mathscr{P}$: Predictors are the set of variables that exist in both DS1 and DS2 that we would like to use to estimate t in DS2.

$\mathscr{O}$: Penalization variables are the set of variables in DS2 that we would like to minimize proxy representation of. In practice, we set $\mathscr{P} \subseteq \mathscr{O}$.

$\mathscr{C}$: Candidates are the set of variables in DS2 we would like to consider as candidates for the recommended proxy.

α: The penalization weight is the hyperparameter ($\in [0,1]$) used to weight the penalty term from candidate proxy correlation with all $v \in \mathscr{O}$.

2.2 Approach

The intuition behind our approach is straightforward: we generate $\hat{t}$ in DS2. These synthetic values represent our best guess for the true values of t; however, we assume that our best guess is flawed. Specifically, we assume that our model has learned a shortcut, and $\hat{t}$ has spuriously high correlations with one or more predictor variables $\in \mathscr{P}$. To ameliorate this issue, `ProxyFinder` calculates a 'Proxy Score' that (1) rewards candidate proxies $c \in \mathscr{C}$ that are strongly correlated with $\hat{t}$, and (2) penalizes candidate proxies that are strongly correlated with the predictor variables $\in \mathscr{P}$. We calculate the positive contribution for (1) for a candidate c as the R^2 value between $\hat{t}$ and c, and the negative contribution (2) as the average R^2 score between c and all predictor variables $\in \mathscr{P}$.

2.3 Algorithm

The `ProxyFinder` algorithm follows a three-stage process, plus a preprocessing step. It can be represented as func(DS1, DS2, t, $\mathscr{P}$, $\mathscr{O}$, $\mathscr{C}$, α) = proxies, where proxies is a list of the top-recommended proxy variables for t in DS2.

To preprocess the two input datasets, we assume that t is observed in DS1 and unobserved in DS2. We assume all variables in $\mathscr{P}$ are observed in both datasets. Finally, we assume that both DS1 and DS2 are large, and sampled from the same population. Mathematically,

$$\forall \mathbf{X_i} \in DS1, \mathbf{X_i} \sim P_i, \text{ and } \forall \mathbf{Y_i} \in DS2, \mathbf{Y_i} \sim P_i$$

Where P_i is the true distribution of variable i. Under this assumption, we are able to use a MinMax scaler to rescale all variable distributions to have a minimum of 0 and maximum of 1 to account for discrepancies in variable coding. For example in our ensuing experiment, a Joe Biden feeling thermometer was coded from 0-5 in DS1 and 0-100 in DS2. Converting these to the same scale led to similar correlations to related variables. In the preprocessing step, we also drop any survey respondents who have at least one unmeasured variable in $\mathscr{P}$ so that we can directly predict t from $\mathscr{P}$ using, for example, a neural network. The first stage of the algorithm is imputation. Without loss of generality, we train an imputation model f to predict t in DS1; i.e., $f(\mathbf{x}^{\mathscr{P}}) = \hat{t}$ for a survey respondent in DS1 $\mathbf{x}$'s values for variables in $\mathscr{P}$. We then similarly use the learned model to infer t in DS2 by using $f(\mathbf{y_i}^{\mathscr{P}}) = \hat{t}$ for all survey respondents in DS2 $\mathbf{y_i}$. If the imputation method is unable to predict t given $\mathscr{P}$ with a MSE of ≥ 0.04, the program terminates[1]. This threshold is slightly arbitrary, and is chosen because the target variable is scaled between [0,1]. `ProxyFinder` is compatible with a range of state-of-the-art imputation methods. We demonstrate the use of two neural network-based imputation methods.

The second stage of the algorithm, and our main contribution, is correlation penalization. Having inferred our predicted values $\hat{t}$ for the target variable in DS2, we would next like to select a candidate proxy that has both (1) a high level of variance explained by the imputed target and (2) a low level of variance explained by a small amount of penalization variables (i.e., to preserve discriminant validity). While there are a few ways to achieve this (as outlined in *Future Directions*), the approach we highlight in this paper is a mean correlation penalty. We calculate a "Proxy Score" for each candidate variable using the following algorithm:

Note that it is also acceptable to use Adjusted R-Squared scores, but unnecessary since we assume large datasets.

In the third stage, we simply select the candidate proxy with the maximal Proxy Score. Mathematically,

[1] This threshold was determined somewhat arbitrarily upon noting a steep performance dropoff around 0.035 training MSE.

Algorithm 1 Proxy Score Calculation

```
scores = []
for c ∈ 𝒞 do
    penalty = 0
    for o ∈ 𝒪 do penalty = penalty + R²_{o∼c} / |𝒪|
    scores[c] = (1 − α) · R²_{f̂∼c} − α · penalty
```

we can represent this algorithm as:

$$\underset{c \in \mathscr{C}}{\operatorname{argmax}} \Big((\frac{1}{\alpha} - 1) \cdot R^2_{\hat{f} \sim c} - \mathbb{M}^q_{o \in \mathscr{O}} [R^2_{o \sim c}] \Big)$$

Where $\mathbb{M}^q$ represents a generalized mean, and in practice we use q=1 (although, as discussed in 'Implications,' this might not be the most effective practice).

3 RESULTS

3.1 Evaluation Metrics

To evaluate the performance of `ProxyFinder` , we use Monte Carlo testing with various predictors and targets in two large political science datasets. We observe model output, and argue for or against the proxy's validity.

3.2 Monte Carlo Simulation for Relevant Penalization Weights

We begin by estimating relevant penalization weights by using one large political science dataset as both the training (DS1) and testing (DS2) datasets. We use data from a 3-wave panel study conducted by YouGov Christopher S. Parker (2020). We create a dataset (YouGov) by isolating a subset of the survey that contains only the target Christian Nationalism and sixteen expert-curated predictor and penalization variables, a full list of which can be found in Appendix A. We then split YouGov into two halves. The first—labeled DS1—was used for training the model, and the second—labeled DS2—was used for imputing the target and selecting a proxy. This experimental design consisted of 50 iterations of Monte Carlo Simulation and two neural network architectures. The first, a Feedforward Neural Network (FFNN), consisted of two hidden layers; each consisted of the ReLu activation function, L2 regularization, and a batch dropout rate of 0.5. The second approach attempts to generate more faithful synthetic values by utilizing a Generative Adversarial Imputation Network (GAIN), based on work by Yoon, Jordon and van der Schaar (2018). Broadly, the GAIN uses a dual-objective approach involving a generator and a discriminator, where the generator attempts to generate realistic data to fool the discriminator, and the discriminator attempts to distinguish the synthetically generated data from real observed data. Hence, it is expected that the GAIN will demonstrate better discriminant validity (i.e., so that the generator can fool

the discriminator), and will not require as large a correlation penalty.

As can be seen in Figure 1, the GAIN performed best with a correlation penalty of 0.63, at which ProxyFinder selected the Christian Nationalism proxy 78% of the time. Conversely, the FFNN performed best with a correlation penalty of 0.68, at which ProxyFinder selected the Christian Nationalism proxy 92% of the time. Further experimentation found the optimal penalty to be highly dependent on the predictive strength of the variables, as well as the target itself (both reflected in the MSE of the model). To capture this variance, we introduce a heuristic; namely, we found a correlation penalty between 0.60 and 0.75 to perform best when using ProxyFinder as described above.

3.3 Monte Carlo Testing of Real Use Cases

To conduct more realistic experimentation, we utilized 15 predictors from YouGov to train the neural network. We used 722 pre-election variables from the American National Election Studies 2020 survey (ANES) to impute the target and select a proxy American National Election Studies (2021). With Biden Favorability as the target, we identified 10 suitable predictors that existed in both datasets, and found that with zero correlation penalty, Trump Favorability was selected 100% of the time under Monte Carlo simulation with 50 iterations each. However, as the correlation penalty was increased to 0.69, a more suitable proxy, namely the Democratic Party Feeling Thermometer (V201156), was selected with 70% frequency. As the correlation penalty increased further, the selection of V201156 declined and instead 'White'— a racial indicator, which assumes the value 1 if the respondent's race is white and 0 otherwise — became the most frequently selected proxy. The full results can be seen in Figure 2.

In similar fashion, with Trump favorability as the target and the same 10 predictor and penalization variables, we found that Biden Favorability (pff_jb) was selected most frequently when using zero correlation penalty (see Figure 3). Specifically, pff_jb was selected with 66% frequency, Immigrant Citizenship with 4% frequency, and the non-predictor variable V201155 (Barack Obama Feeling Thermometer) with 30% frequency. As the penalty rose, V201155 became the most selected proxy with 70% frequency at a correlation penalty of 0.72. Notably as the correlation penalty increased past 0.80, an arguably more suitable proxy of V201410 (Transgender Policy), began to be selected. These results are contrasted with prior experimentation on the truncated YouGov survey. Specifically, weak proxies, such as the respondent's ID, dominated the selection when penalties greater than 0.75 were applied.

While the above results are promising, selecting a proxy for other variables, such as Abortion (ANES code: V201336) or Immigrant Deport (ANES code: V202245x), proved much more difficult. As can be seen in Figure 4, at the optimal correlation penalties, the number of unique proxies selected by the algorithm exceeded 20, when run with just 50 iterations per penalty. Even without any correlation penalty, 5 unique proxies were selected for Abortion, all of which were themselves predictor/penalization variables.

Even when removing variables that could be considered redundant proxies, the variance in selection remained high. For example, at a correlation penalty of 0.70, the most frequent proxies for Abortion were V201503 (Financially Better Off), V201215 (Republican Candidate Honest), and Immigrant Citizenship, which is a predictor/penalization variable. None of these proxies were selected more than 5% of the time, and convergence toward a singular variable was only achieved at much higher penalties.

4 DISCUSSION AND IMPLICATIONS

Our testing has shown that ProxyFinder selects a non-penalization variable with moderate success in divergent datasets (such as the YouGov and the ANES 2020 survey). As demonstrated by experimentation on political candidates' feeling thermometers, ProxyFinder is able to select our face-valid proxy with $\geq 70\%$ accuracy for penalties 0.69 ± 0.015.

When plotted as a function of correlation penalty, the proxy selection frequency exhibits several peaks. As shown in Figures 2 and 3, a predictor/penalization variable (pff_jb and pff_dt, respectively) is selected between 0 and approximately 0.65, which we argue is due to the strong correlation between the imputed target and the features used in the model. While some may consider pff_jb and pff_dt to be suitable proxies, we explicitly reject this notion by including them in the list of penalization variables. Consequently, without a correlation penalty, simply predicting the target in the second dataset would not suffice.

In contrast, the second peak in both Figures 2 and 3 represents the "natural" proxy, one whose correlation with the predictor variables is suppressed but not to the extent that it loses predictive power or fails to capture sufficient variance. V201156 (the Democratic Party Feeling Thermometer) is one such example (as a proxy for the Joe Biden Feeling Thermometer target variable). Respondents in the ANES survey were asked to enter a number between 0-100 to rate their opinion of the Democratic Party following the national convention (the survey was conducted between August 18, 2020 and November 3, 2020). As such, Joe Biden would be viewed as the unambiguous leader of the party, and V201156 would serve as an excellent proxy for Biden favorability. As the selection of V201156 decreases with $\alpha \geq 0.70$, the binary racial variable, White, begins to dominate. We suspect that when correlation with the predictor variables is suppressed too heavily, weak proxies like race, or even nonsensical proxies like the respondent's ID are selected. As demonstrated in the table below, we effectively better capture the variance of our target (Biden favorability) with the second peak (Democratic party favorability) rather than the first peak (Donald Trump Favorability).

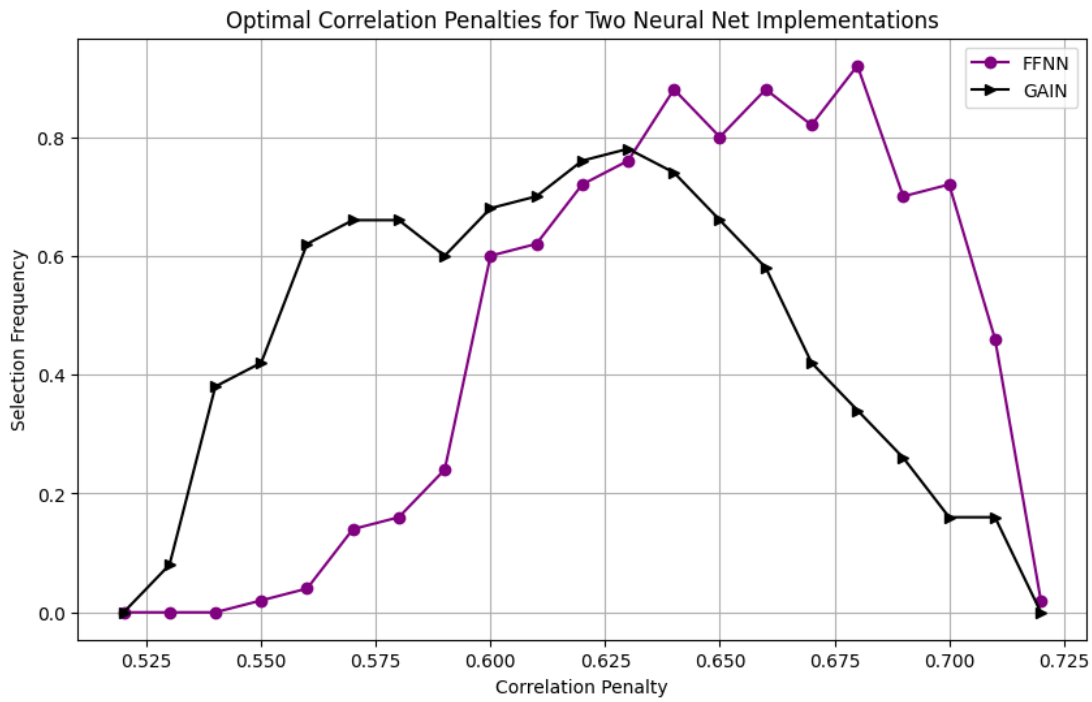

Figure 1. Proxy selection as a function of correlation penalty for two neural network architectures, the feed forward network and the generative adversarial imputation network. The target was Christian Nationalism in the YouGov.

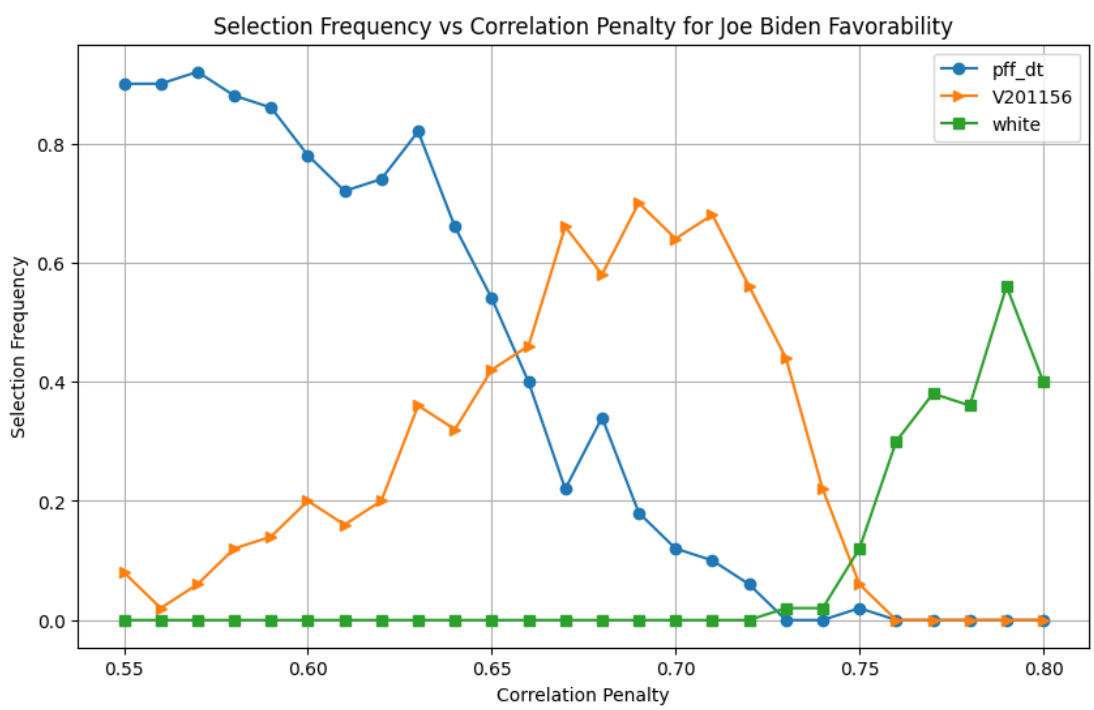

Figure 2. Proxy selection as a function of correlation penalty for Biden Favorability. The variables pff_dt, V201156, and white refer to Trump Favorability, The Democratic Party Feeling Thermometer, and the White racial variable, respectively.

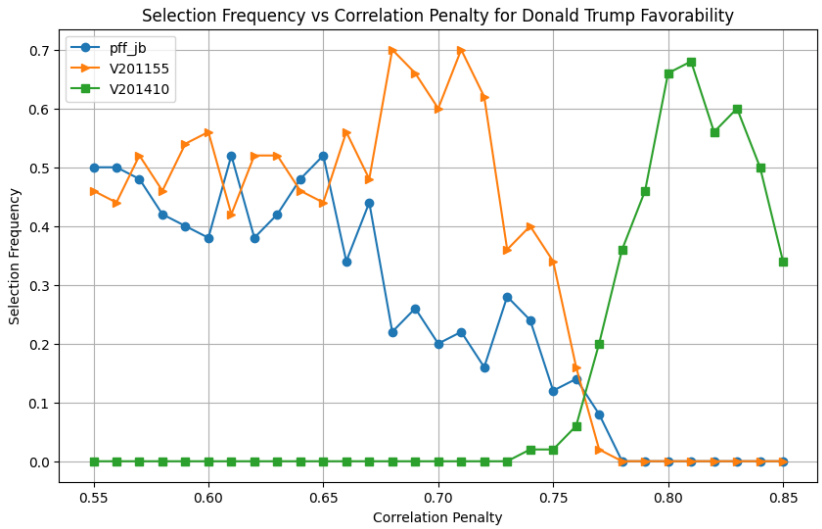

Figure 3. Proxy selection as a function of correlation penalty for Trump Favorability. The variables pff_jb, V201155, and V201410 refer to Biden Favorability, the Barack Obama Feeling Thermometer, and Transgender Policy, respectively.

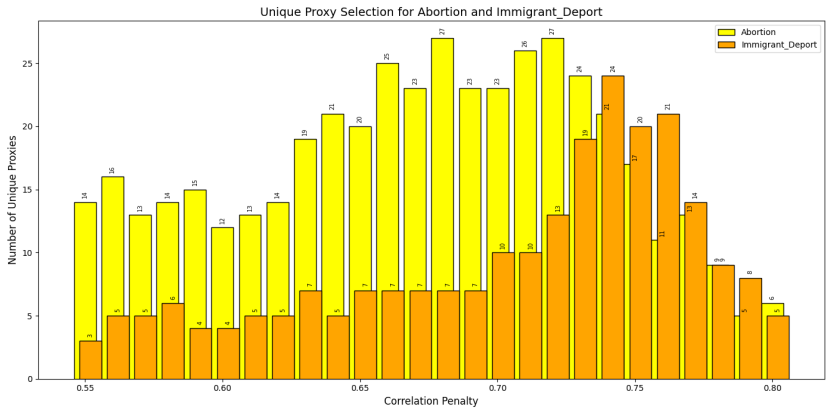

Figure 4. Unique proxy selection as a function of correlation penalty for the Abortion and Immigrant Deport target variables.

	pff_jb	**immigrant_deport**	**pff_dt**	**abortion**
pff_dt	1.0	0.88	0	0.86
immigrant_citizenship	0	0.12	0.04	0
pff_jb	0	0	0.66	0.06
V201155	0	0	0.30	0
ideo7	0	0	0	0.04
auth_grid_2	0	0	0	0.02
immigrant_deport	0	0	0	0.02

Table 1. Proxy selection for four target variables with zero correlation penalty. V201155 (Feeling Thermometer: Barack Obama) was the only non-predictor variable selected, with 0.30 frequency for the target pff_dt.

	pff_jb (target)	pff_dt (peak 1)	V201156 (peak 2)	White (peak 3)
R^2 with pff_jb	1.0	0.65	0.74	0.04

Table 2. Three peak correlations with target pff_jb. The variables pff_dt, V201156, and white refer to Trump Favorability, The Democratic Party Feeling Thermometer, and the White racial variable, respectively.

The same periodic pattern appears when using Trump Favorability as a target variable, now with V201155 (Barack Obama Feeling Thermometer) as the second peak. At first glance, V201155 may not appear as a suitable proxy for Trump Favorability, but the natural alternative—the Republican Party Feeling Thermometer—would be of little use. Supporters of Donald Trump have historically held mixed views of the Republican Party, with many deriding the party's willingness to compromise on key pieces of legislation, or denouncing its leadership as the establishment. Similarly to our previous experiment, the third peak (V201410 - Transgender Policy) explains very little variance of our target, and should be disregarded.

	pff_dt (target)	pff_jb (peak 1)	V201155 (peak 2)	V201410 (peak 3)
R^2 with pff_dt	1.0	0.65	0.67	0.02

Table 3. Three peak correlations with target pff_dt. The variables pff_jb, V201155, and V201410 refer to Biden Favorability, Obama Favorability, and Transgender Policy, respectively.

We suspect that the optimal correlation penalty, and the location of the aforementioned peaks, highly depends on the accuracy of the neural network. We observe that as the training mean squared error (MSE) $\rightarrow 0$, $\alpha \rightarrow 0$. Conversely, as train MSE $\rightarrow \infty$, $\alpha \rightarrow \infty$[2]. This is reflected in the data, as the average MSE was 0.0337, 0.0348, 0.0361, and 0.0374 for Biden Favorability, Trump Favorability, Abortion, and Immigrant Deport, respectively. We observed that the MSE fluctuated significantly for different targets, making it difficult to determine a precise optimal correlation penalty. Instead, we introduced the heuristic described above, namely that a correlation penalty between 0.60 and 0.75 performs best when using ProxyFinder.

During Monte Carlo testing of the algorithm, we observed significant fluctuations in the model's mean squared error due to the inherent randomness in the initialization of weights and biases. These fluctuations resulted in inconsistent proxy rankings, with certain variables shifting by as many as 10-15 positions in large datasets like the ANES. To minimize this issue, we suggest running ProxyFinder for a minimum of 70 iterations per correlation penalty. Another approach could be to truncate the dataset and explicitly remove redundant variables, such as summary metrics. Finally, for replicability and stability,

[2]Note that we cannot observe inference MSE as the target variable is not measured in DS2

the user can initialize the weights with a random seed.

Another source of MSE variation was due to the target itself. Abortion and Immigrant Deport were particularly hard to predict, with the MSE 0.0361 and 0.0374, respectively. Because the error term was significantly higher than Donald Trump and Joe Biden Favorability, not only was the optimal penalty much higher but the number of unique proxies selected per weight was quite large. As can be seen in Figure 4, the algorithm generated more than 20 unique proxies for certain correlation penalties. Some of these were redundant proxies, but even removing similar metrics had little impact. We suspect that as the MSE increases, convergence to a singular proxy, or even a handful of proxies, may become unachievable. In particular, some of the most obvious proxies for Abortion, namely explicit questions about abortion, were rarely selected, which reinforces the notion that the target may not have been accurately predicted.

5 FUTURE DIRECTIONS

5.1 Classifiers

Improving the prediction of the target using architectures other than a FFNN or GAIN could improve proxy selection. We suspect that with categorical or discrete variables, which comprise most of the ANES dataset, a classification model would yield better performance. If the two Feeling Thermometers (V201155 and V201156, coded on 100-point scales in ANES) are thought of as continuous variables, it comes as no surprise that the MSE for these predictions was marginally lower than for the Abortion or Immigrant Deport targets (V201336 or V202245x, coded on 4- and 7-point scales).

5.2 Nonlinear Correlations

We should also consider proxies that have strong nonlinear relationships with the target. In the future we will explore, e.g., polynomial regressions or another neural network with a MSE metric.

5.3 Generalized Means

Our objective function uses the standard mean $\mathbb{M}^1$ to penalize a candidate proxy's ability to explain the penalization variables. However, this does not theoretically achieve what we want it to. Our goal is to penalize candidate proxies that have a large amount of their variance explained by a small amount of the predictor variables (i.e., to preserve discriminant validity). Because of this, using a generalization of the mean $\mathbb{M}^q$ with $q > 1$ would more accurately represent our objective.

ACKNOWLEDGMENTS

We would like to thank Princeton University's Office of Undergraduate Research for enabling this research through the Hewlett Foundation Fund.

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A PREDICTOR VARIABLES IN INITIAL MONTE CARLO TESTING

The following variables were used as both predictor and penalization variables in the initial Monte Carlo testing:

- **Presvote20post** – Respondent's vote for President in 2020
- **Housevote20post** – Respondent's vote for the House of Representatives in 2020
- **Senatevote20post** – Respondent's vote for the Senate in 2020
- **Pff_jb** – Favorability rating of Joe Biden
- **Pff_dt** – Favorability rating of Donald Trump
- **Pid7** – Party identification (7-point scale)
- **Election_fairness** – Confidence in the fairness of the 2020 presidential election
- **Educ** – Highest level of education attained
- **Hispanic** – Racial indicator variable for Hispanic identity
- **Partisan_violence** – Belief that sending threatening or intimidating messages to party leaders is acceptable
- **Immigrant_citizenship** – Support (7-point scale) for a path to citizenship for unauthorized immigrants who otherwise obey the law